PHYS-1310
Quiz 1
Given Time: 1 hr 15 min

Name: _____ Student ID: _____

INSTRUCTION: Choose the best alternative and write it on the separate answer sheet.

PART I: Multiple Choice question 1-24 (Conceptual - each question worth 1 point= 24%)

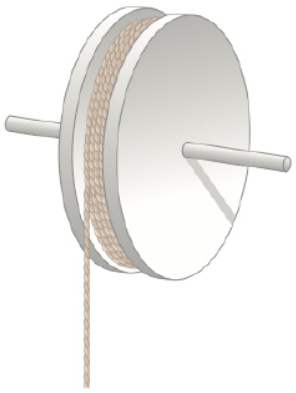
1.A particle travels along a curved path between two points P and Q as shown. The displacement of the particle does *not* depend on

A) the location of P.
B) the location of Q.
C) the distance traveled from P to Q.
D) the shortest distance between P and Q.
E) the direction of Q from P.

2. The driver of a speeding empty truck slams on the brakes and skids to a stop through a distance d. On a second trial, the truck carries a load that doubles its mass. What will now be the truck's "skidding distance"?
   (a) $4d$
   (b) $2d$
   (c) $\sqrt{2}\, d$
   (d) d
   (e) $d/2$
3. If an object is in equilibrium, which of the following statements is *not* true?
   (a) The speed of the object remains constant.
   (b) The acceleration of the object is zero.
   (c) The net force acting on the object is zero.
   (d) The object must be at rest.
   (e) There are at least two forces acting on the object.
4. A truck loaded with sand accelerates along a high- way. The driving force on the truck remains constant. What happens to the acceleration of the truck if its trailer leaks sand at a constant rate through a hole in its bottom?
   (a) It decreases at a steady rate.
   (b) It increases at a steady rate.
   (c) It increases and then decreases.
   (d) It decreases and then increases.
   (e) It remains constant.
5. A large crate of mass m is place on the flatbed of a truck but not tied down. As the truck accelerates for- ward with acceleration a, the crate remains at rest relative to the truck. What force causes the crate to accelerate?
   (a) the normal force
   (b) the gravitational force
   (c) the friction force
   (d) the ma force exerted by the crate
   (e) No force is required.
6. Consider an object on a rotating disk a distance r from its center, held in place on the disk by static friction. Which of the following statements is *not* true concern ing this object?
   (a) If the angular speed is constant, the object must have constant tangential speed.
   (b) If the angular speed is constant, the object is not accelerated.

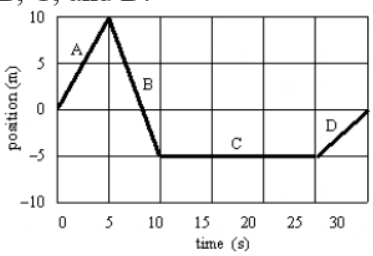
(c) The object has a tangential acceleration only if the disk has an angular acceleration.
(d) If the disk has an angular acceleration, the object has both a centripetal acceleration and a tangential acceleration.
(e) The object always has a centripetal acceleration except when the angular speed is zero.

7. A constant net torque is exerted on an object. Which of the following quantities for the object cannot be constant? Choose all that apply.
   (a) angular position
   (b) angular velocity
   (c) angular acceleration
   (d) moment of inertia
   (e) kinetic energy
8. A solid aluminum sphere of radius R has moment of inertia I about an axis through its center. Will the moment of inertia about a central axis of a solid aluminum sphere of radius $2R$ be
   (a) $2I$,
   (b) $4I$,
   (c) $8I$,
   (d) $16I$, or
   (e) $32I$
9. As shown in Figure, a cord is wrapped onto a cylindrical reel mounted on a fixed, frictionless, horizontal axle. When does the reel have a greater magnitude of angular acceleration?
   (a) When the cord is pulled down with a constant force of 50 N.
   (b) When an object of weight 50 N is hung from the cord and released.
   (c) The angular accelerations in parts (a) and (b) are equal.
   (d) It is impossible to determine.



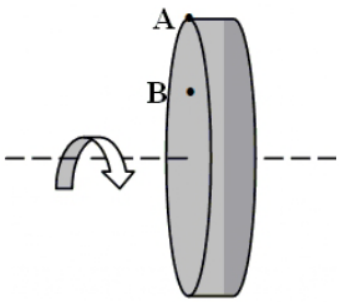
For problems 10-11

An object is moving along the x axis. The graph shows its position from the starting point as a function of time. Various segments of the graph are identified by the letters A, B, C, and D.



10. What is the velocity of the object at $t = 7.0$ s?
    A) +3.0 m/s
    B) –1.0 m/s
    C) –2.0 m/s
    D) –3.0 m/s
    E) zero m/s

11. What is the acceleration of the object at $t = 7.0$ s?
   A) zero m/s^2
   B) –2.0 m/s^2
   C) –3.0 m/s^2
   D) +9.8 m/s^2
   E) +4.0 m/s^2
12. The mass of a raindrop is 10 milligrams. Which one of the following statements indicates the correct mass of the raindrop in grams?
   A) The raindrop has a mass of 10^6 grams.
   B) The raindrop has a mass of 10^{-2} grams.
   C) The raindrop has a mass of 10^{-1} grams.
   D) The raindrop has a mass of 10^3 grams.
   E) The raindrop has a mass of 10^{-6} grams.
13. An object moving along a straight line is decelerating. Which one of the following statements concerning the object's acceleration is *necessarily* true?
   A) The value of the acceleration is positive.
   B) The direction of the acceleration is in the same direction as the displacement.
   C) An object that is decelerating has a negative acceleration.
   D) The direction of the acceleration is in the direction opposite to that of the velocity.
   E) The acceleration changes as the object moves along the line.
14. A rigid body rotates about a fixed axis with a constant angular acceleration. Which one of the following statements is true concerning the tangential acceleration of any point on the body?
   A) The tangential acceleration is zero m/s^2.
   B) The tangential acceleration depends on the angular velocity.
   C) The tangential acceleration is equal to the centripetal acceleration.
   D) The tangential acceleration is constant in both magnitude and direction.
   E) The tangential acceleration depends on the change in the angular velocity.
15. Which one of the following statements concerning a wheel undergoing rolling motion is true?
   A) The angular acceleration of the wheel must be zero m/s^2.
   B) The tangential velocity is the same for all points on the wheel.
   C) The linear velocity for all points on the rim of the wheel is non-zero.
   D) The tangential velocity is the same for all points on the rim of the wheel.
   E) There is no slipping at the point where the wheel touches the surface on which it is rolling.
16. Two points are located on a rigid wheel that is rotating with *decreasing* angular velocity about a fixed axis. Point A is located on the rim of the wheel and point B is halfway between the rim and the axis. Which one of the following statements concerning this situation is true?



   A) Both points have the same centripetal acceleration.
   B) Both points have the same tangential acceleration.
   C) The angular velocity at point A is greater than that of point
   D) Both points have the same instantaneous angular velocity.
   E) Each second, point A turns through a greater angle than point

17. A baseball is hit upward and travels along a parabolic arc before it strikes the ground. Which one of the following statements is necessarily true?
    A) The acceleration of the ball decreases as the ball moves upward.
    B) The velocity of the ball is zero m/s when the ball is at the highest point in the arc.
    C) The acceleration of the ball is zero m/s^2 when the ball is at the highest point in the arc.
    D) The x-component of the velocity of the ball is the same throughout the ball's flight.
    E) The velocity of the ball is a maximum when the ball is at the highest point in the arc.
18. A solar-powered car is traveling at constant speed around a circular track. What happens to the centripetal acceleration of the car if the speed is doubled?
    A) The centripetal acceleration remains the same.
    B) The centripetal acceleration increases by a factor of 2.
    C) The centripetal acceleration increases by a factor of 4.
    D) The centripetal acceleration is decreased by a factor of one-half.
    E) The centripetal acceleration is decreased by a factor of one-fourth.
19. A ball is whirled on the end of a string in a horizontal circle of radius R at constant speed v. Complete the following statement: The centripetal acceleration of the ball can be increased by a factor of 4 by
    A) keeping the speed fixed and increasing the radius by a factor of 4.
    B) keeping the radius fixed and increasing the speed by a factor of 4.
    C) keeping the radius fixed and increasing the period by a factor of 4.
    D) keeping the radius fixed and decreasing the period by a factor of 4.
    E) keeping the speed fixed and decreasing the radius by a factor of
20. Which one of the following situations is *not* possible?
    A) A body has zero velocity and non-zero acceleration.
    B) A body travels with a northward velocity and a northward acceleration.
    C) A body travels with a northward velocity and a southward acceleration.
    D) A body travels with a constant velocity and a time-varying acceleration.
    E) A body travels with a constant acceleration and a time-varying velocity.
21. An object moving along a straight line is decelerating. Which one of the following statements concerning the object's acceleration is *necessarily* true?
    A) The value of the acceleration is positive.
    B) The direction of the acceleration is in the same direction as the displacement.
    C) An object that is decelerating has a negative acceleration.
    D) The direction of the acceleration is in the direction opposite to that of the velocity.
    E) The acceleration changes as the object moves along the line.
22. Which of the following is NOT an example of accelerated motion

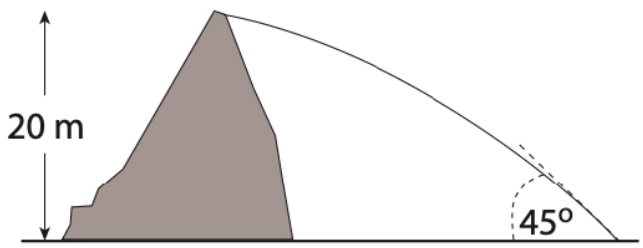
    a) Horizontal component of a projectile motion
    b) Vertical component of a projectile Motion
    c) Circular motion at a constant speed
    d) A swinging pendulum
    e) Earths motion about the sun

23. Does a car moving around a circular track with constant speed have
    (a) zero acceleration,
    (b) an acceleration in the direction of its velocity,
    (c) an acceleration directed away from the center of its path,
    (d) an acceleration directed toward the center of its path, or
    (e) an acceleration with a direction that cannot be determined from the given information.

24. An astronaut hits a golf ball on the Moon. Which of the following quantities, if any, remain constant as a ball travels through the vacuum there?
    (a) speed
    (b) acceleration
    (c) horizontal component of velocity
    (d) vertical component of velocity
    (e) velocity

PART II: 18 problems — 25-43 (4 points each= 76%)

25. Let A, B and C be any vectors. Which of the following statements is false?
    a) $A \cdot B = B \cdot A$
    b) $A+(B+C) = (A+B) + C$
    c) $A \times (B \times C) = (A \times B) \times C$
    d) $A \cdot (B+C) = A \cdot B + A \cdot C$
26. Which vector is not equal to zero?
    a) $i \times (j \times k)$
    b) $(i \times j) \times k$
    c) $(i \times i) \times j$
    d) $i \times (i \times j)$
27. Let $A = a_1 i + a_2 j + a_3 k$ and $B = b_1 i + b_2 j + b_3 k$. Then $(A \times B) \cdot j$ is equal to
    a) $a_2 b_3 - a_3 b_2$
    b) $a_3 b_1 - a_1 b_3$
    c) $a_1 b_2 - a_2 b_1$
    d) $a_1 b_3 - a_3 b_1$
28. A ball rolls up a slope. At the end of three seconds, its velocity is 20 cm/s; at the end of eight seconds, its velocity is 0 cm/s. What is the average acceleration (in cm/s^2) from the third to the eighth second?
    (a) 4.0
    (b) 2.5
    (c) 5.0
    (d) 6.0
    e) 6.67
29. The coordinate of an object is given as a function of time by $x = 7t - 3t^2$, where x is in meters and t is in seconds. Its average velocity over the interval from t = 0 to t = 2 s is:
    (a) −5 m/s
    (b) 1 m/s
    (c) 11 m/s
    (d) −11 m/s
    (e) −14.5 m/s
30. A stone is thrown horizontally from the top of a 20-m high hill. It strikes the ground at an angle of 45°. With what speed was it thrown?
    a. 20 m/s
    b. 14 m/s
    c. 28 m/s
    d. 32 m/s
    e. 50 m/s

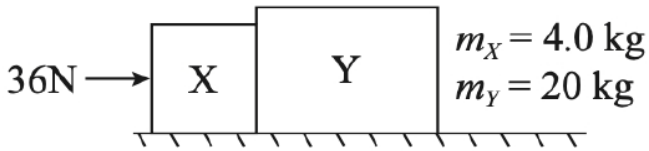


31. Two forces are applied to a 5.0-kg object; one is 6.0 N to the north, and the other is 8.0 N to the west. The magnitude of the acceleration of the object is:

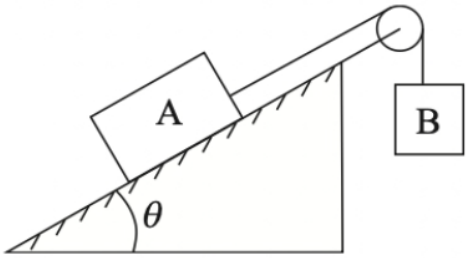
(a) 2.0 m/s^2
(b) 0.50 m/s^2
(c) 2.8 m/s^2
(4d 10 m/s^2
(e) 50 m/s^2

32. Two blocks (X and Y) are in contact on a horizontal frictionless surface. A 36-N constant force is applied to X as shown. The force exerted by Y on X is:

a) 30N
b) 1.5 N
c) 6.0 N
d) 29N
e) 36N

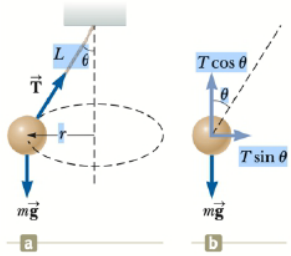


33. Block A, with a mass of 10 kg, rests on a θ = 30° incline. The coefficient of kinetic friction is 0.20. The attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. Block B, with a mass of 8.0 kg, is attached to the dangling end of the string. The acceleration of B is:



a) 0.69 m/s^2, down
b) 0.69 m/s^2, up
c) 2.6 m/s^2, down
d) 2.6 m/s^2, up
e) 0

34. A small ball of mass *m* is suspended from a string of length *L*. The ball revolves with constant speed *v* in a horizontal circle of radius *r* as shown in Figure (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum.*) Find an expression for *v* in terms of the geometry in



(a) $\boxed{v = \sqrt{Lg\ sin\theta tan\theta}}$

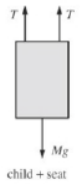
(b) $\boxed{v = \sqrt{rg}}$

(c) $\boxed{v = rgcos\theta}$

(d) $\boxed{v = rgsin\theta}$

35. (a) A child of mass M swings in a swing supported by two chains, each of length R. If the tension in each chain at the lowest point is T, find what is the child's speed at the lowest point.

a) $v = \sqrt{(2T - Mg)\left(\frac{R}{M}\right)}$
b) $v = 2T\text{-}Mg$
c) Mv^2/R
d) None



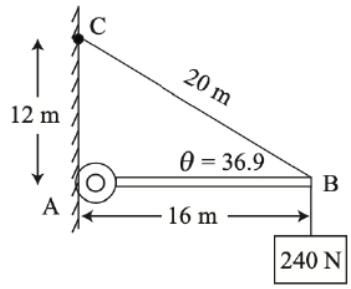
36. A disk with a rotational inertia of 5.0 kg·m^2 and a radius of 0.25 m rotates on a frictionless fixed axis perpendicular to the disk and through its center. A force of 2.0 N is applied tangentially to the rim. If the disk starts at rest, then after it has turned through half a revolution, its angular velocity is:

(a) 0.80 rad/s
(b) 0.57 rad/s
(c) 0.64 rad/s
(d) 1.6 rad/s
(e) 3.2 rad/s

37. A 240-N block is suspended as shown. The beam AB is weightless and is hinged to the wall at A. The tension in the cable BC is:

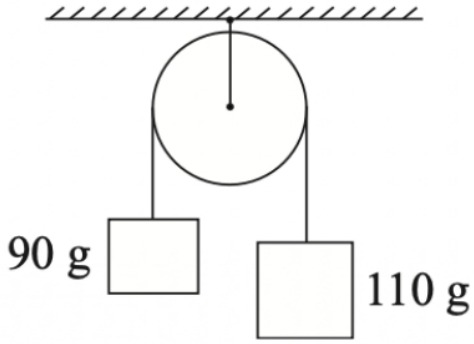
(a) 400 N
(b) 300 N
(c) 320 N
(d) 180 N

(e) none of these



38. Two blocks are connected by a string and pulley as shown. Assuming that the string and pulley are massless, the magnitude of the acceleration of each block is:

(a) 1.0 m/s^2
(b) 0.05 m/s^2
(c) 0.02 m/s^2
(d) 0.1 m/s^2
(e) 10 m/s^2



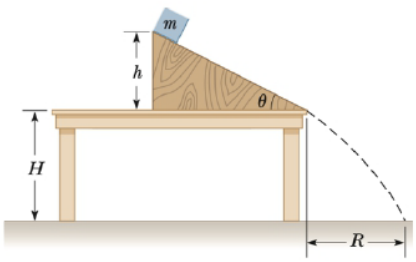
39. A high tower is built on planet X (which has no atmosphere and has an acceleration due to gravity of 8 m/sec^2). A man standing at the top of the tower leans over the edge and simultaneously drops two balls one up and one down each with an initial speed of 5 m/s. What is the distance between the balls 3 seconds later?
    a) 30m
    b) 36m
    c) 18m
    d) 72m
    e) 24m
40. A disk 8.00 cm in radius rotates at a constant rate of M 1 200 rev/min about its central axis. the tangential speed at a point 3.00 cm from its center, and the radial acceleration of a point on the rim, are
    (a) 3.77 m /s and 1260m/s^2
    (b) 126 m /s and 1700m/s^2
    (c) 3.77 m /s and 1260m/s
    (d) 3.77 m /s^2 and 1260m/s^2
    (e) 3.77 rad /s and 1260m/s^2

41. A certain light truck can go around a curve having a radius of 150 m with a maximum speed of 32.0 m/s. To have the same acceleration, at what maximum speed can it go around a curve having a radius of 75.0 m?
    (a)64m/s
    (b)45m/s
    (c)32m/s
    (d)23m/s
    (e)16m/s

42. A block of mass m_1 = 2.00 kg and a block of mass m_2 = 6.00 kg are connected by a massless string over a pulley in the shape of a solid disk having radius R = 0.250 m and mass M =10.0 kg. The fixed, wedge- shaped ramp makes an angle of θ = 30.0° as shown in Figure. The coefficient of kinetic friction is 0.360 for both blocks. The acceleration of the two blocks is

    a) 0.609m/s^2
    b) 0.309 m/s^2
    c) 3.09 m/s
    d) 30.9m/s

43. A block of mass m = 2.00 kg is released from rest at h = 0.500 m above the surface of a table, at the top of a θ = 30.0° incline as shown in Figure. The frictionless incline is fixed on a table of height H = 2.00 m. How far from the table will the block hit the floor?



    a) 0.135 m
    b) 2.0m
    c) 13.5m
    d) 1.35m

PHYS-1310
Quiz 1 Answer Sheet

Name: _____ Student ID: _____

Part I: 1 point each

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____
17. _____
18. _____
19. _____
20. _____
21. _____
22. _____
23. _____
24. _____

Part II: 4 Points each

25. _____
26. _____
27. _____
28. _____
29. _____
30. _____
31. _____
32. _____
33. _____
34. _____
35. _____
36. _____
37. _____
38. _____
39. _____
40. _____
41. _____
42. _____
43. _____